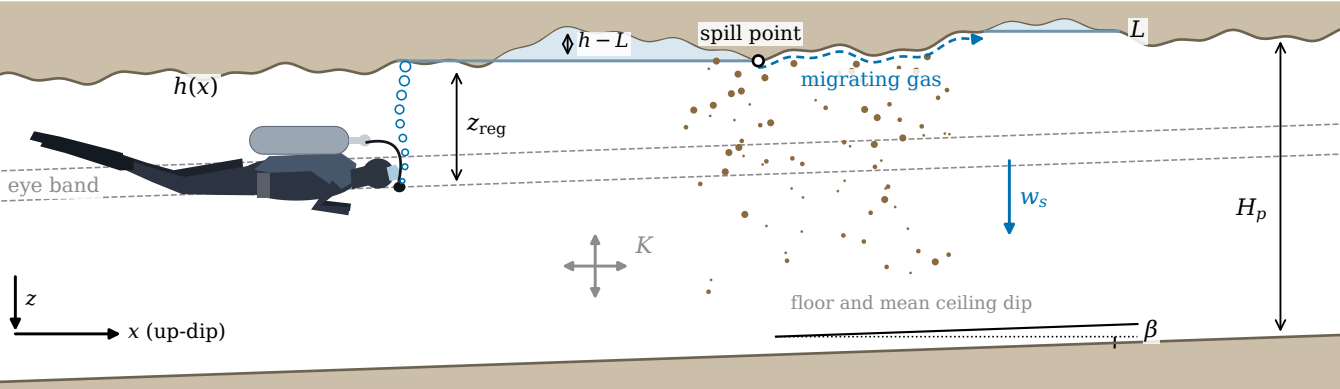
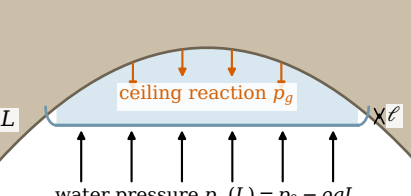
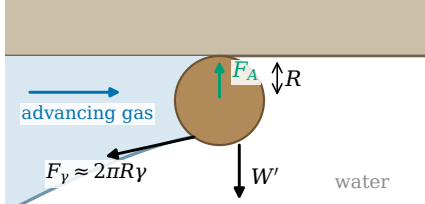


a) Model section along the passage, to scal

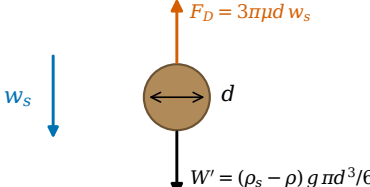
**(b) Gas pool**

$\ell = 2v/(\rho q r)$ pins the rim

(c) Grain under a contact line

detaches when $F_v > F_A + W'$

(d) Settling grain



$$W' = F_D \Rightarrow w_s = (\rho_s - \rho) q d^2 / (18\mu)$$